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Review

Extreme events impact attribution: A state of the art

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https://doi.org/10.1016/j.crsus.2024.100101

SUMMARY

Extreme weather events lead to many adverse societal, economic, and environmental consequences. Anthropogenic climate change has been identified as a factor that may have already increased the frequency and intensity of these events. The methods of extreme event attribution (EEA) that quantify the extent to which climate change affected specific recent extreme weather events in the past 20 years have been combined more recently with socio-economic impact data to quantify extreme weather's impacts attributable to climate change. This extreme event impacts attribution (EEIA) is a quickly developing field that considers which kinds of questions about the impacts of climate change on extreme weather events we should ask, what methods are best suited to answer them, how to interpret the results these methods provide, and what purpose these results can serve. We discuss the methods of EEIA, review EEIA results, and discuss their implications and potential uses.

INTRODUCTION

Extreme weather events lead to many wide-ranging adverse consequences, disrupting many aspects of social and cultural life, the economy, and the natural environment. Climate change, caused by the anthropogenic emission of greenhouse gases (GHGs), has been identified as a factor that is changing the frequency and intensity of weather extremes in many locations and, in most cases, increasing both. These influences, however, differ across extreme weather types (e.g., extreme high temperatures, extreme precipitation) and regions. This link between extreme weather events and anthropogenic climate change is most evident for heatwaves but has also been quantified for extreme precipitation events, droughts, tropical cyclones, and compound events such as fire weather.[1]

Various links between climate change and long-term weather trends have been previously described, with the attribution of long-term trends to climate change being investigated even before the first Intergovernmental Panel on Climate Change (IPCC) Report in 1990.[2] Yet, a different approach—examining how anthropogenic climate change affected a specific weather event, in terms of its meteorological characteristics—is a more recent endeavor. A method for attribution of individual extreme weather events was first proposed by Allen,[3] who described how to assess the extent to which external drivers (e.g., GHGs) changed the occurrence probability of a specific extreme weather event based on climate modeling. Since the earliest application of this approach for the deadly continental European heatwave of 2003,[4] the methods of extreme event attribution (EEA) have evolved considerably, and their application in hundreds of research projects now spans a much wider set of extreme weather types and geographical regions. These research projects enable researchers to estimate how anthropogenic climate change affected certain characteristics of extreme weather events such as temperature, precipitation (intensity and duration), or wind speed.

The applications of EEA have typically been limited to studying how climate change affected the meteorological characteristics of the weather event, without considering how those changing meteorological characteristics translate to social, economic, or environmental impacts. More recently, however, EEA has begun to be employed in combination with information about the impacts of extreme weather events, to link climate change to its impacts on socio-economic systems through the changes it makes to extreme weather. Attribution has yet to be employed for environmental or ecological impacts—see Lloyd and Shepherd.[5] We refer to the efforts to attribute socio-economic impacts as extreme event impacts attribution (EEIA), following the terminology suggested in Perkins-Kirkpatrick et al.[6] This strand of research now has the potential to add to the other existing methods of analyzing climate change impacts, such as the ones based on attribution of impacts to long-term climate trends[7,8] or those based on integrated assessment models (IAMs).

EEIA is a recent and rapidly developing field, and there is a significant debate over what kinds of questions about the impacts of climate change on extreme weather events can be answered with EEIA, what methods are best suited to answer them, how to interpret the results these methods provide, and what purpose these results can serve. In the next sections, we (1) discuss the basic structure and methods of EEIA, (2) review the results of

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the existing EEIA studies, and (3) discuss the implications and outlook for this strand of research.

IMPACTS ATTRIBUTION METHODS

EEIA studies employ standard EEA methods and extend their application using socio-economic impact data to quantify the contribution of climate change to the observed impact of extreme weather events. The EEIA "algorithm" can be divided into the EEA part (the climate science component) and the impact analysis (the socio-economic component, including potentially human and environmental impacts). The EEA component is used to determine the change in the probability or the intensity of a well-defined extreme weather event (e.g., the rainfall in a particular location and period) as a consequence of climate change. The impact part then links this change to the event's observed impacts (e.g., mortality associated with the event) to attribute a fraction of the total impact of the event to anthropogenic climate change causes. In this review, we focus on the general documentation of impacts of human-induced climate change, so we do not consider the role of non-climate causal factors—specifically exposure and vulnerability to hazard events—in impact outcomes. We do not consider how their effect is modulated by climate change, nor how they may modulate climate change.

EEA

Impact attribution of extreme weather is primarily based on the effect of climate change on the trigger of these impacts—the weather hazard itself; it does not probe the effects of climate change on exposure and vulnerability. EEA is typically based on the comparison of climate model simulations run under conditions with and without anthropogenic forcings. This can be done in various forms. Approaches are often classified as following a probabilistic approach (also referred to as the risk-based approach) and a storyline approach (also referred to as the mechanistic or process-based approach). The probabilistic approach, which originates from Allen,[3] conceptualizes an individual extreme weather event as a member of a certain class of events (typically defined by an exceedance of a certain threshold, e.g., a level of precipitation) and attempts to estimate how an external driver (typically anthropogenic GHGs) influenced the probability of this class of events occurring.[9] In an alternative framing, this probability approach can also be used to estimate how climate change altered the intensity of a class of events with a specific return period.[10] Both of these framings involve comparing event probabilities or intensities in the present world (i.e., with climate change) and a counterfactual world where no climate change is occurring and atmospheric GHG levels are in their pre-industrial levels. Probability-based assessments are usually based on climate modeling but can also be based on extreme value statistics of observational or reanalysis climate data, which uses historical climate in which human influence is assumed to be small to estimate the counterfactual weather.[11] There is an active debate within the climate attribution scientific literature about the relative merits of these two approaches. As the focus in this survey is on impact attribution, we choose not to adjudicate on this point here.

The usage of storyline is an alternative to the probabilistic approach in that it decomposes the event into various component processes and diagnoses the anthropogenic role in at least some of those processes. A frequent approach is to distinguish the dynamic (specific weather patterns) from the thermodynamic (e.g., surface warming or atmospheric moistening) factors responsible for an extreme weather event. Although the probabilistic approach attempts to simulate both dynamic and thermodynamic factors together, the storyline approach often takes the dynamic conditions preceding an event as given[12] (i.e., dynamically conditioned attribution). This approach then attempts to quantify the contribution of the anthropogenic forcings to the event's thermodynamic variables, focusing on the change in the event's intensity and disregarding considerations of probability. In practice, this dichotomy is imperfect, and many EEA studies employ elements from both methodologies. The methods used may often not be easily categorized as they differ mostly in the degree of conditioning.[13,14]

The EEA estimates are influenced by many factors, such as the framing of the attribution question, the definition of the event (choice of the variable being analyzed, spatial and temporal extent of the defined event), and the methodologies used.[15,16]

For the purposes of impact attribution (EEIA), it is important that the event is defined by variables that are closely linked to the event's impacts, spatially and temporally, and in terms of the analyzed variables—for example, wind speed vs. rainfall in a tropical storm.[17] Often, the extreme weather event is characterized by a single weather variable, and this needs to align with the variable that causes most of the impact—for example, for most tropical cyclones, most of the damage and almost all of the fatalities are typically caused by the associated rainfall rather than the wind, even when wind speed is often used to characterize the intensity of a cyclone.[18]

This can be a limitation in cases where the impacts are driven by compound events (e.g., a drought event caused by a combination of high temperature, high wind, and a lack of rainfall), as the role of some of these contributing factors and the climate change component of these may be overlooked.[14] EEA estimates are also necessarily associated with a degree of uncertainty related to the role of natural variability, the use of imperfect climate models, statistical sampling, and other factors.[19]

In addition to these caveats, there is a considerable geographical imbalance in the existing EEA analyses, as they are mostly conducted for weather events in the high-income countries. This geographic imbalance arises because of the availability of longer and better historical data, well-evaluated regional climate models, and research funding.[20]

Attribution impact studies employ various EEA methods spanning both the probabilistic and storyline approaches. Some studies rely on existing EEA estimates from previous research,[21–26] while other studies include EEA analyses as part of their contribution.[6,27–38]

Translating EEA to impacts = EEIA

Once the influence of climate change on a specific extreme weather event has been estimated with EEA, this information can be used to determine the contribution of climate change to the event's impacts. A basic methodological distinction in the

existing EEIA studies lies in whether the impact is attributed based on an EEA estimate of the change in the probability of the event or its intensity. In all cases, however, the working ceteris paribus assumption is that while the hazard has changed, because of climate change, exposure and vulnerability have not (i.e., they are "held constant"). As such, all EEIA papers ignore the possibility that anthropogenic climate change has also changed the exposure of assets and people to the hazard and their vulnerability to it.

The earliest proposal on how to attribute impacts from extreme weather events to climate change was based on the changes in the event's probability. Allen[3] stated that: "If at a given confidence level, past greenhouse gas emissions have increased the risk of a flood 10-fold, and that flood occurs, then we can attribute, at that confidence level, 90% of any damage to those past emissions" (p. 2). This approach assumes that the relative contribution of climate change to the frequency of a particular class of events is equal to the attributable fraction of impacts resulting from a single event from that class. This approach typically employs the fraction of attributable risk (FAR) measure, which estimates what percentage of a certain class of events have occurred due to climate change and is based on a comparison of probabilities in contemporary and counterfactual simulated climates.[39] To calculate an event's attributable impact, the FAR estimate is multiplied by the event's impact. We henceforth refer to this method as IFAR (IFAR = impacts*FAR), following Brown.[40] The IFAR approach has been applied to attribute extreme weather's economic impacts,[22–24,30,31] or to attribute mortality from heat waves[21,32] and hurricanes.[21] In some of these cases (e.g., Mitchell et al.[32]) the FAR estimate is based on a quantity from the impact system derived from further modeling on top of the climate modeling rather than on a relevant climate variable.

An alternative to this approach is EEIA based on the attributed change in the event's intensity. The intensity is similarly estimated for both factual and counterfactual climate conditions, so that a counterfactual intensity (without climate change) is calculated and compared with the actual intensity of the event—for example, in terms of precipitation. These different estimated intensities are then used with a damage function to calculate the event's estimated impacts in the counterfactual case of no climate change and compared with the event's realized impacts. The attributable impact is then calculated as the difference between the event's impacts under the factual and counterfactual conditions.

Unlike with the IFAR approach, this intensity-based approach always requires a (non-step) damage function.[6] Intensity-based EEIA studies use various types of "damage functions," which link a meteorological or hydrological variable to an impact category of interest. For example, some heatwave-related mortality/morbidity studies use existing temperature-mortality/morbidity functions, parameterized based on historical data, to estimate heat-related deaths or morbidity for both factual and counterfactual simulated heatwave scenarios.[6,33,35–37]

Another strand of intensity-based studies employs hydrological models to translate precipitation into flood-extent maps for estimation of the number of flooded properties[29,34] and to estimate the economic damages from these floods by constructing three-dimensional flood depth maps and matching these with damage functions that define building damage as a function of flood depth.[25,38] Both Villarini et al.[38] and Smiley et al.[25] utilize these depth-damage functions, while Wehner and Sampson[26] use a simpler functional form connecting spatial flood extent to damage linearly. Using an alternative approach, Perkins-Kirkpatrick et al.[6] employ previously established damage functions directly linking precipitation to financial damages in the form of excess insurance payouts based on previously estimated insurance liability functions from Pástor-Paz et al.[40]

Probability vs. intensity in EEIA

The influence of climate change on a single weather event may appear different depending on whether a change in probability or intensity is examined. Due to the estimated shape of the probability distributions, relatively minor changes in intensity may result in large changes in probability.[41] This has been illustrated, for example, in the case of a Russian heatwave where climate change was estimated to lead to minor changes in the event's intensity but a major increase in the frequency (probability of occurrence) of such an event.[10] Bearing the difference between intensity and probability in mind, the seeming difference between intensity- and probability-based weather attribution statements appears to be reconcilable.

But the issue seems to be different for impacts attribution. Event attribution analysis addresses the anthropogenic role in a specific event.[42] If that specific event is defined by its probability, then the attribution ought to follow a probability-based framework. In most cases, though, an impact event is defined by its intensity (e.g., the financial cost of damage or fatalities), and so the attribution ought to follow an intensity-based framework.[43] The recent COVID epidemic illustrated this distinction (Dann Mitchell, pers. comm.). Much of the public discourse surrounded the excess mortality during the pandemic: this was a question of intensity—how much the number of fatalities had increased because of the spread of the new pathogen. However, public policy was more concerned about the risk of failure of the public health system, such as the chance that hospital capacity would be exceeded: this was a question of probability of exceedance of a threshold due to the spread of the pathogen.

When it comes to EEIA the difference in the results arrived at by using the two different methods may be significant, as was demonstrated for the case of Hurricane Harvey. The total attributable economic damages were estimated to be at 75% of total damages (USD 67 billion) using the probability approach,[23] and a much lower estimate of 14% (USD 13 billion) of total damages using the intensity-based approach.[26] The attributable damages to residential properties were estimated at 37% and 58% for the lower and higher climate scenarios, respectively, also using the intensity-based approach.[25] The difference between these results is not only because of their reliance on different EEA methods. The papers make very different simplifying assumptions about the spatial distribution of damage and the relevant damage functions. Much of the difference is related to these differences rather than the probability/intensity dichotomy.

In a direct comparison of methods, Perkins-Kirkpatrick et al.[6] found important differences in the results derived from probability-based vs. intensity-based methods for two case studies

looking at mortality and insurance payouts. Nevertheless, the resources required for EEIA mean that it may often be the case that only one method is feasible, even if that method is less appropriate from a technical point of view. For instance, intensity-based approaches require a model of the impact system—i.e., a damage function that is more nuanced than a binary step function. The development of a new impact model may be beyond the resources allocated to attribution research, and thus obtaining an attribution assessment is only feasible with methodological compromises. In that case, it is useful to understand how transferable results are across methods.[6,16,26,44]

The accuracy of the damage functions that are assumed or estimated is greatly relevant to the quality of the results. Deployment of reliable damage functions may be limited by factors such as a lack of relevant data to estimate their functional forms. This constraint applies especially to low- and middle-income countries, as micro-level impact accounting is not as prevalent. For example, Chapman et al.[27] attempt to attribute child mortality impacts in Africa to climate change but must rely on existing heat-mortality studies from only two locations. From a temporal perspective, some studies (e.g., Vicedo-Cabrera et al.[36]) use a constant damage function for EEIA over a longer period, which may fail to account for the adaptive changes being made. Stuart-Smith et al.[35] employ a time-varying damage function, to better accommodate for adaptation. They show that the use of a constant heat-mortality function underestimated heat-related mortality in the beginning of the estimated time span and overestimated it in the later period.

The availability and quality of impact data for the investigated events also represent a limiting consideration, as EEIA studies must rely on existing estimates of the events' impacts (or on proxies for these that require their own set of sometimes tenuous assumptions). Incomplete event impact data is clearly a limiting factor for the attribution of certain kinds of impacts (e.g., psychological damage) or for some geographical regions that have fewer data collection capacities (e.g., low-income countries). This problem is especially pertinent to attributing the indirect impacts of events (e.g., business interruption losses, disruptions of supply chains), which are already much more difficult to quantify and are rarely included in disaster impact assessments.[45] Consequently, many EEIA studies that focus on economic impacts only consider direct or insured damages.

IMPACT ATTRIBUTION FINDINGS

What were the results of the existing EEIA studies? The two main strands of studies attribute, first, human impacts such as human mortality, morbidity, and displacement, and second, economic impacts, such as direct and indirect damages, insured losses, or the number of flooded properties. A handful of studies attribute both human and economic impact. A full list of the EEIA studies we identified is presented in Table 1, including the information on whether probability- or intensity-based method is used, the type of event, the type of attributed impact, and the country/region of the study. Some studies attempt to attribute extreme weather impacts to climate change without employing formal EEA methods, and we include some of these studies (denoted by *) in Table 1 as well.

The majority of the EEIA studies focusing on human impacts analyze heat-related mortality. Mitchell et al.[32] is the first study to apply the probability method in an EEIA context. Incidentally, it analyzes the same event that was the focus of the first EEA study, the European heatwave in the summer of 2003. It estimates that the effect of climate change accounted for 20% of heat-related deaths in London and 69% in Central Paris during that heatwave. Since damage functions of mortality as a function of temperature can easily be constructed from past data, further research into the attribution of heat-related mortality largely focused on the intensity (e.g., Vicedo-Cabrera et al.[36,37]).

In the same context, Perkins-Kirkpatrick et al.[6] attribute heat-related mortality for the hottest day during the record-breaking 2006 UK summer. They take on the point also made by Brown,[40] which is critical of the probability approach, between a class of events and an event of a specific intensity. They estimate the average return period of an event with an impact equal to the observed one and compare this impact with the impact of a counterfactual event with the same return period. They suggest that the attributable impact is different when considering a climate change signal behind a class of events (37%–50% of deaths are attributable for a class of events leading to at least 60 deaths) compared with a specific event (17% of deaths are attributable when an event leading to 60 deaths occurs).

Another pair of papers that investigate the same event using both approaches are Frame et al.[23] and Smiley et al.[25] The first attribute the direct economic cost of Hurricane Harvey in 2017 using averages of the available fractions of attributable risk and estimates of the economic damage wrought by the hurricane. Their results suggest that 75% of the economic cost (approximately USD 67 billion) was attributable to climate change. Smiley et al.,[25] by contrast, use valuation data and depth-damage functions to model the damage of the actual and the counterfactual floods for each building in the city. When using an estimate of 38% precipitation attributable to climate change, they calculate that about 50% of the flooded properties were flooded because of climate change (i.e., would not have flooded otherwise), and accounting for approximately 60% of the total property damages.

A completely different question is asked by Lott et al.[31] They extend the probabilistic methodology to analyze how a portion of the attributed cost can be assigned to individuals based on their historical carbon emissions as it is approximated by their age and nationality. Using a case study of a summer heatwave in Eastern China in 2018, the authors calculate that sample individuals are responsible for between 0.53 and 18.10 yuan of the event's impact on aquaculture, with the value changing with their age and their country's emission profile.

HOW USEFUL IS EEIA

EEIA presents a bottom-up approach to estimate the costs of climate change that are caused by the change in frequency and intensity of extreme weather events. It is an alternative to the top-down approach of integrated assessment modeling (IAM). The IAM community has been developing, during the last few decades, a whole suit of modeling approaches that aim to comprehensively assess the cost of climate change and

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Table 1. EEIA studies

Impact	EEIA method	Study	Event type	Type of attributed impact	Country/region
Human impact	probability	Mitchell et al.[32]	heatwave	mortality	UK
	intensity	Vicedo-Cabrera et al.,[36,37] Stuart-Smith et al.,[35] and Åström et al.[46]	heatwave	mortality	43 countries (Vicedo-Cabrera et al.[36]), Switzerland (Vicedo-Cabrera et al.[37] and Stuart-Smith et al.[35]), and Sweden (Oudin Åström et al.[46])
		Chapman et al.[27]	heatwave	child mortality	Africa
		Puvvula et al.[33]	heatwave	morbidity	USA
		Mester et al.[47]	hurricane	displacement	Mozambique
Economic impact	probability	Frame et al.[22]	flood and drought	direct insured cost (floods) and indirect loss (droughts)	New Zealand
		Frame et al.[23] and Li and Otto[30]	hurricane	direct cost (Frame et al.[23]) and insured loss (Li and Otto[30])	USA (Frame et al.[23]) and Japan (Li and Otto[30])
		Lott et al.[31]	heatwave	losses to aquaculture	China
	intensity	Wehner and Samson[26] and Strauss et al.[48]	hurricane	direct cost	USA
		Smiley[25] and Villarini et al.[38]	hurricane (Smiley[25]) and flood (Villarini et al.[38])	property damage	USA
		Schaller et al.[34],* and Kay et al.[29]	flood	number of flooded properties	UK
Human and economic impact	probability	Dalagnol et al.[28]	flood	direct and indirect costs and injuries/displacement/ homelessness	Brazil
		Clarke et al.[21]	heatwave, flood, and hurricane	mortality (heatwave), direct cost (flood), direct cost, and mortality (hurricane)	UK (heatwave, flood) and Puerto Rico (hurricane)
		Newman and Noy[24]	heatwaves, floods, droughts, wildfires, and storms	direct cost and mortality cost	worldwide
	intensity	Perkins-Kirkpatrick et al.[6]	heatwave and flood	mortality (heatwave) and insured loss (flood)	UK (heatwave) and New Zealand (flood)

thus inform mitigation policy—by, for example, identifying the social cost of carbon. Yet, the IAMs have been criticized for their necessarily simplified modeling approaches, which, according to some critics, render their quantifications useless (e.g., Pindyck[49]). A large part of the expected cost of climate change is mediated through extreme weather events. One application of EEIA is therefore to compare those estimates of extreme weather-related costs from IAMs and the estimates obtained from EEIA. Newman and Noy[24] do that for the most well-known IAM: the DICE model, whose most recent iteration is described in Barrage and Nordhaus.[50]

IAMs have their own advantages, of course; for example, they are less reliant on the idiosyncratic disaster damage and loss data collection practices that are apparent in many countries,[51] nor the various methodological concerns we discussed with EEIA. In addition, the IAMs' attempts to quantify the costs of climate change are more comprehensive (include other types of costs such as those associated with slow-onset changes in the climate).

Another potential use for EEIA is the quantification of climate loss and damage,[52] and especially as these quantifications pertain to the discussions about the Loss and Damage Fund (L&D Fund) within the United Nations Framework Convention on Climate Change. The plan to establish an L&D Fund dates back more than a decade, to the 19th Conference of the Parties (COP 19) held in Warsaw in 2013, but the L&D Fund was only officially established in the 28th COP in 2023.[53] EEIA could become one of the main sources of quantified information that will inform the ways in which money will be allocated to the L&D Fund by the source countries, and money from the Fund will be provided to the "vulnerable developing countries" that are to be the Fund's recipients. King et al.[54] suggest that, currently, event attribution should only play a limited role in determining L&D spending due to the existing methodological issues and disparities within the

field pertaining to geographical representation and process understanding of various hazard types. However, Noy et al.[55] argue that EEIA can already assist with quantifying allocations for very impactful extreme events in vulnerable countries.

In addition to the "compensation" that may be provided through the L&D Fund and which EEIA can quantify, EEIA may also have the potential to inform climate change litigations. The most obvious use of EEIA in climate change litigation would be to provide a metric for awarding compensation. However, much of the present climate litigation originates from individuals (or non-governmental groups) suing governments and corporations, mostly for their inaction on mitigation. As classified by Setzer and Higham,[56] these legal cases typically ask either the government or corporations to consider climate change in setting their policies, to improve their enforcement of their existing climate-friendly policies, to make them channel more resources, to improve their adaptation practices, and to prevent them from engaging in climate-washing. Only a small minority of cases are about obtaining compensation. According to Setzer and Higham,[56] out of the 230 cases they identified as strategic, only 11 dealt with compensation at all, and in 2 of them, compensation was only a subsidiary issue. The most well-known compensation case is Luciano Lliuya vs. RWE AG, a case that was filed on behalf of a Peruvian farmer against the largest German electricity company in 2015 and whose appeal is being considered by the Higher Regional Court in Germany. However, even this case is not about a specific extreme weather event that has already happened, so EEIA would not be useful for quantification of liabilities in this case.

By itself, however, EEA could be important for legal challenges, as it might allow plaintiffs to overcome one important legal hurdle that had tripped previous cases. As noted by Setzer and Vanhala,[57] some important climate change cases were dismissed because the issue was deemed non-justiciable (as the chain of causality from emissions to extreme weather was seen as unproven). EEA provides that causal chain, but EEIA could go a step beyond this, and provide plaintiffs with the assessment of the amount of damage for which they may be entitled to. As noted by Jézéquel et al.,[58] one of the main challenges for EEA's inclusion in legal arguments is that it only examined the hazard (the natural phenomenon) rather than the risk of an event and the way it materialized in damages. EEIA provides that alternative lens.

Another potential use of EEIA is as a source of insight for planning and prioritizing climate change adaptation. Most obviously, adaptation requires scarce resources, and those resources should first be directed to where they can provide the most social benefit. In other words, *ceteris paribus*, adaptation funding should be directed to those places where the damage from climate change is already high or predicted to be high soon, and the cost of adaptation is low. EEIA could help in determining the former, while careful adaptation planning can inform the latter (e.g., Litzow et al.[59]).

One additional use of EEIA is in raising public awareness of climate risks.[60] EEA, to some extent, is already providing this increased recognition of the link between anthropogenic GHG emissions and experienced extreme weather disasters, especially through such work as done by the World Weather Attribution Network.[17] However, similarly to the potential role of EEIA in litigation, EEIA can provide an easier-to-grasp and more intuitive metric for the connection between GHG and disaster impacts, measured in lives and dollars, rather than in changing probabilities of events (e.g., Lott et al.[31]).

Another issue, already mentioned earlier, is the *ceteris paribus* assumption with respect to exposure and vulnerability. Of course, changes in exposure and vulnerability influence the impact of an extreme event (that point has been recognized for decades in the disaster risk literature). Climate change may also have an impact on exposure and/or vulnerability and thus change the impact of an event not only by altering the hazard profile (which is what attribution estimates do). The impact of an event can also be reduced. This reduction can be achieved not only by changing the hazard profile (through GHG mitigation, for example) but also by reducing exposure and/or vulnerability. This possibility, however, is irrelevant to the use cases suggested above. Indeed, once you start unspooling the connections to exposure and vulnerability, it becomes apparent that it is always the case that climate change is not the only cause of an impact, in much the same way as smoking is not the only cause for lung cancer. As such, accounting for exposures and vulnerabilities is largely irrelevant to the questions surrounding loss and damage and liability.

THE FUTURE OF EEIA

EEIA has only been implemented in the past few years, and many methodological challenges still need to be resolved and some scientific disagreements need to be adjudicated. Further progress in the methods of event attribution, including the potential for ultra-high-resolution climate modeling, could reduce the uncertainty associated with attribution statements regarding localized extremes such as storms, hail, or tornadoes, and improve our ability to make attribution statements on compound events.[54]

Still, the data for impacts are still lacking in many respects.[21,61] Some types of extreme weather events are covered much less than others. For example, there is little data on the impact of heat waves in low- and middle-income countries,[51] and much more data on the impact of tropical cyclones in high-income countries. Some regions are covered much better than others as well (generally there is much less data on both mortality and economic impacts in low-income countries—Jones et al.[62]).

These data often cover only direct damages, however, and do not include the indirect losses to economic activity and human health, and consequently to societal well-being. A full inventory of impacts must also account for these indirect losses, but these are much more challenging to measure consistently; they even lack a consensus definition or terminology.

The EEA data are equally incomplete, with most attribution studies focused on events in high-income countries, with some geographical regions having no attribution studies on certain recurring types of extreme events.[24] Clarke et al.,[52] however, provide a road map for using existing EEA to inform the attribution of events for which an EEA is not available. Their logic clearly also applies to EEIA.

Better and more complete data on impacts, more attribution analyses, and further development of the EEIA attribution methods and their extrapolation to other events are all required so that EEIA can fulfill the needs of a global community that is trying to deal with the challenge posed by anthropogenic climatic change.

ACKNOWLEDGMENTS

I.N., D.S., and T.U. were supported by the New Zealand Ministry of Business, Innovation, and Employment through the Whakahura Endeavour project.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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